SYNAPSE EDGE DEFENSE

Schema Learning Lifecycle

How Synapse infers per-endpoint JSON schemas from live traffic — and why training is deferred until after the body-phase WAF verdict. Microsecond validation. No manifests. Poison-resistant baselines.

~5µs	~3µs	1.5×	LRU
LEARN PER REQUEST	VALIDATE PER REQUEST	LENGTH TOLERANCE	SCHEMA EVICTION

REQUEST PATH — VALIDATE NOW, TRAIN LATER

STAGE 01
Parse & Template
JSON body parsed. Path normalized to an endpoint template (e.g. /api/users/:id).
OBSERVE

STAGE 02
Validate Against Schema
Existing schema (if any) checks field types, patterns, lengths, bounds. Emits ValidationResult.
INFER

STAGE 03
Body-Phase WAF
Rules consume the validation result via schema_violation match kind alongside DLP & SQLi checks.
WAF

STAGE 04
Verdict Gate
Allow or block. Training payload stashed in pending_learn remains untouched until this gate passes.
GATE

STAGE 05
Train If Allowed
On allow: consume pending_learn, update field types & bounds. On block: drop it — baseline untouched.
APPLY

→ ALLOW PATH
SCHEMA_LEARNER.learn_from_request(template, body) fires. The sample increments the endpoint's baseline.

→ BLOCK PATH
pending_learn is dropped without consumption. Attack payloads never enter the baseline.

WHAT THE LEARNER CAPTURES

Field Types
PRIMITIVE CLASSIFICATION
string · number · boolean
null · object · array
mixed (ambiguity flag)
Promotes to mixed on divergence

Value Patterns
SEMANTIC INFERENCE
UUID · email · ISO date
URL · IP · hex digest
regex-detected formats
Via profiler::patterns::detect_pattern

Length & Value Bounds
NUMERIC CONSTRAINTS
min / max string length
min / max numeric value
tolerance multiplier applied
Default tolerance: 1.5× learned max

Sample Count
TRAINING MATURITY
Per-endpoint sample counter
min_samples_for_validation
Silent until matured
Prevents false positives while learning

Nested Structure
OBJECT TREES
Recursive field maps
Array element schemas
Depth-limited traversal
max_nesting_depth guards memory

Endpoint Identity
TEMPLATE KEY
Keyed by normalized path
LRU eviction past max_schemas
DashMap thread-safe
Generation queue for O(1) eviction

EXAMPLE — POST /API/USERS

OBSERVED REQUEST BODY

```
{
  "id": "3f1a9c7e-2b4d",
  "email": "alice@ex.com",
  "age": 34,
  "verified": true,
  "joined": "2026-04-12"
}
```

→

INFERRED FIELD SCHEMA

```
// endpoint: /api/users · samples: 42
id       → uuid    (len 36)
email    → email   (len ≤ 48)
age      → number  (0–120)
verified → boolean
joined   → iso-date
```

DOWNSTREAM CONSUMERS OF THE SCHEMA

WAF · schema_violation
BODY-PHASE RULE KIND
Rules match on total_score via compare_threshold (gte/gt/eq/lte/lt). Evaluated synchronously in the body phase — same pass that produces the verdict.

DLP Scanner
POST-VERDICT HAND-OFF
Learned field patterns inform which fields are candidates for PII / secret scanning. Runs after the WAF verdict on allowed bodies.

Anomaly Detection
STRUCTURAL DRIFT
Unexpected fields, missing required fields, and type promotions to mixed surface as anomalies — feeding risk scores and Horizon telemetry.

Anti-poisoning guarantee
The body-phase WAF runs before the learner trains. If a request is blocked — SQLi, schema violation, DLP hit — its payload is dropped without entering the baseline. Attackers cannot teach Synapse to trust the shape of an attack, even under sustained traffic. Schemas stay clean.

Schemas that learn from good traffic only.
Microsecond inference. Deferred training. No manifests, no manual tuning, no poisoning surface.